

3.2: Quadratic functions

Definition.

A quadratic is a function of degree two. The **general form of a quadratic function** is

$$f(x) = ax^2 + bx + c \quad (a \neq 0)$$

where a , b , and c represent constants. The **standard form of a quadratic** is

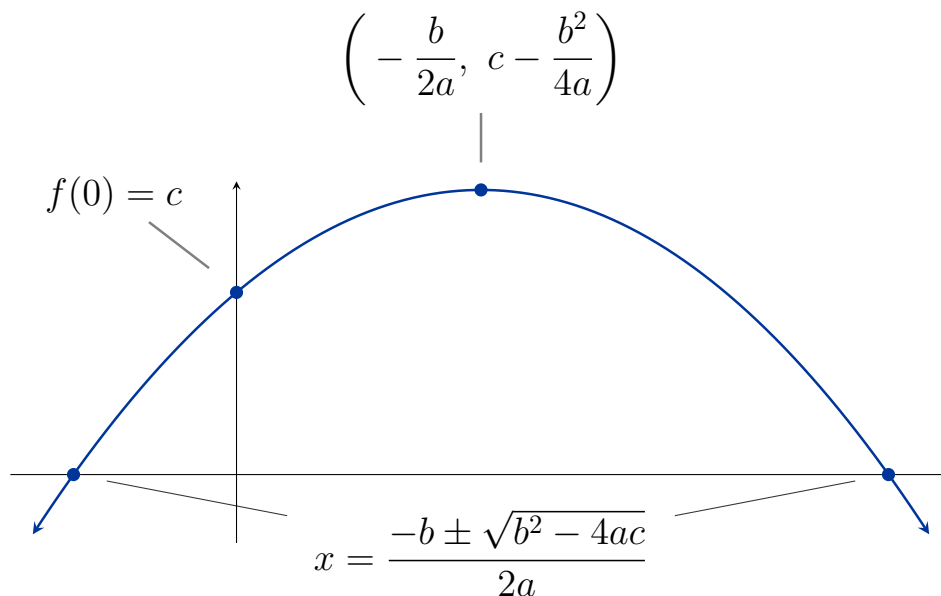
$$f(x) = a(x - h)^2 + k.$$

A **parabola** is the shape of the graph of a quadratic function, with a vertex at (h, k) :

$$h = -\frac{b}{2a}, \quad \text{and} \quad k = f(h) = c - \frac{b^2}{4a}$$

The optimal value occurs at the vertex of a parabola:

- A maximum if $a < 0$ 
- A minimum if $a > 0$ 



Example. Expand the standard form of a quadratic, then write h and k in terms of a , b , and c .

Example. Find the vertex of $f(x) = 2x^2 - 6x + 7$ by

- computing h and k , and
- rewriting the quadratic in standard form by completing the square.

Identify if the vertex is the maximum or minimum value.

- Factor “ a ” from the first two terms

- Add and subtract h^2

- Bring $-h^2$ out

- Simplify

Example. Find the vertex of $g(x) = 13 - 3x^2 + 4x$ by

- computing h and k , and
- rewriting the quadratic in standard form by completing the square.

Identify if the vertex is the maximum or minimum value.

- Factor “ a ” from the first two terms

- Add and subtract h^2

- Bring $-h^2$ out

- Simplify

Example. For the following quadratic functions, find

- the x and y -intercepts,
- the vertex, and
- the domain and range.

$$f(x) = 3x^2 + 2x - 4$$

$$g(x) = -(x + 2)^2 - 1$$

Solving quadratic equations via factoring:

Example. Solve $2x^2 + x = 3x + 12$

1. Rewrite the equation in the general form: $2x^2 - 2x - 12 = 0$

2. Rewrite bx using factors of ac : $2x^2 - 6x + 4x - 12 = 0$

3. Factor out like terms: $2x(x - 3) + 4(x - 3) = 0$

4. Factor by grouping: $(x - 3)(2x + 4) = 0$

5. Solve for the roots: $x = 3$ and $x = -2$

Example. Solve the following for x via factoring:

$$(x + 3)(x - 1) = 5$$

$$-4x^2 + 8x - 3 = 0$$

Example. A backyard farmer wants to enclose a rectangular space for a new garden within her fenced backyard. She has purchased 80 feet of wire fencing to enclose three sides, and she will use a section of the backyard fence as the fourth side (see below). What dimensions should she use to maximize the enclosed area?

